Symbols used in 'dirac matrices' and their definitions

Every symbol of the proof file 'dirac matrices' is listed with its defining equation.

symbol	meaning	definition
$x_0, \dots, x_7$	Cartesian coordinates of the project	x_0 hidden space, x_1, x_2, x_3 3-space, x_4 time, x_5, x_6, x_7 superluminal time
A, B, C, D	frame indices	$A, B, C, D \in \{0, 1, \dots, 7\}$
η^{AB}, η_{AB}	flat metric of signature (4,4)	$\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$
δ_{pq}	Kronecker delta	$\delta_{pq} = 1$ if $p = q$, else 0
ϵ_{pqrs}	Levi-Civita symbol	totally antisymmetric, $\epsilon_{1234} = 1$
I_n	identity matrix	$n \times n$ unit matrix
σ	8x8 spinor metric	$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$
Q_a^h, Q_b^h	4x4 building blocks	$(Q_a^h)_{pq} = \epsilon_{hpq4}$, $(Q_b^h)_{pq} = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$, $h = 1, 2, 3$
s_h	self-dual antisymmetric 4x4	$s_h = Q_a^h - Q_b^h$, $\frac{1}{2}\epsilon_{pqrs}(s_h)_{rs} = (s_h)_{pq}$
t_h	anti-self-dual antisymmetric 4x4	$t_h = Q_a^h + Q_b^h$, $\frac{1}{2}\epsilon_{pqrs}(t_h)_{rs} = -(t_h)_{pq}$
τ_A	real 8x8 matrices	$\tau_0 = I_8$, $\tau_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}$, $\tau_{3+j} = \begin{pmatrix} 0 & t_{4-j} \\ -t_{4-j} & 0 \end{pmatrix}$, $\tau_7 = \tau_1 \cdots \tau_6$
$\bar{\tau}_A$	partner matrices	$\bar{\tau}_A = \sigma \tau_A^T \sigma$; $\frac{1}{2}(\tau_A \bar{\tau}_B + \tau_B \bar{\tau}_A) = \eta_{AB} I_8$
$T^A = (T16^A)[A]$	the eight real 16x16 Dirac matrices	$T^A = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$, $T^A T^B + T^B T^A = 2\eta^{AB} I_{16}$
$\Gamma = (T16^A)[8]$	chirality matrix	$\Gamma = T^0 T^1 \cdots T^7 = \mathrm{diag}(-I_8, I_8)$, $\Gamma T^A = -T^A \Gamma$
P_L, P_R	chiral projection matrices	$P_L = \frac{1}{2}(I_{16} - \Gamma)$, $P_R = \frac{1}{2}(I_{16} + \Gamma)$
σ_{16}	16x16 spinor metric (Dirac adjoint)	$\sigma_{16} = T^0 T^1 T^2 T^3 = \mathrm{diag}(-\sigma, \sigma)$; $\bar{\psi} = \psi^\dagger \sigma_{16}$
S^{AB}	scaled commutators (Lie algebra generators)	$S^{AB} = \frac{1}{4}(T^A T^B - T^B T^A)$
L^{AB}	so(4,4) generator on vectors	$(L^{AB})^C{}_D = \eta^{BC}\delta^A_D - \eta^{AC}\delta^B_D$
$D^{AB}_{(1)}$, $D^{AB}_{(2)}$	type-1 and type-2 spinor generators	$4D^{AB}_{(1)} = \bar{\tau}_A \tau_B - \bar{\tau}_B \tau_A$, $4D^{AB}_{(2)} = \tau_A \bar{\tau}_B - \tau_B \bar{\tau}_A$, $S^{AB} = \mathrm{diag}(D^{AB}_{(1)}, D^{AB}_{(2)})$
θ	group parameter	real number
$\rho(g)$, $\Lambda(g)$	(twisted) adjoint action on vectors	$\rho(g) T^C = (-1)^{\lvert g\rvert} g T^C g^{-1} = \Lambda(g)^C{}_D T^D$
$\lvert g\rvert$	parity of a Clifford element	0 for products of an even number of vectors, 1 for odd
Cl(4,4)	real Clifford algebra	generated by e^A with $e^A e^B + e^B e^A = 2\eta^{AB}$; Cl(4,4) = M(16, R)
unit vector	element $v = v_A e^A$	$v \cdot v = \eta^{AB} v_A v_B = \pm 1$
Pin(4,4)	group of products of unit vectors	double cover of O(4,4) via ρ
Spin(4,4)	even part of Pin(4,4)	products of an even number of unit vectors
O(4,4), $\mathrm{SO}^+(4,4)$	orthogonal groups	$L\eta L^T = \eta$; SO^+ = identity component
so(4,4)	Lie algebra of O(4,4)	real 8×8 X with $X\eta + \eta X^T = 0$; dimension 28
B_I	basis of M(16,R)	$B_I = T^{A_1} \cdots T^{A_k}$, $I = \{A_1 < \cdots < A_k\}$, $B_\emptyset = I_{16}$; $\mathrm{Tr}(B_I B_J^{-1}) = 16\delta_{IJ}$
c, s	coefficients in Lemma 4	$(c, s) = (\cos\frac{\theta}{2}, \sin\frac{\theta}{2})$ or $(\cosh\frac{\theta}{2}, \sinh\frac{\theta}{2})$; $e^{\theta S^{AB}} = cI_{16} + 2sS^{AB}$
X	auxiliary matrix in the proof of Theorem 2	$X = \tau_A^{-1}\tau_B$, $X^{-1} = -\eta_{AA}\eta_{BB}X$ for $A \neq B$
ψ, $\bar{\psi}$	16-component spinor and its adjoint	$\bar{\psi} = \psi^\dagger \sigma_{16}$; $\bar{\psi}\psi$ is Spin(4,4) invariant because $(\sigma_{16} S^{AB})^T = -\sigma_{16} S^{AB}$